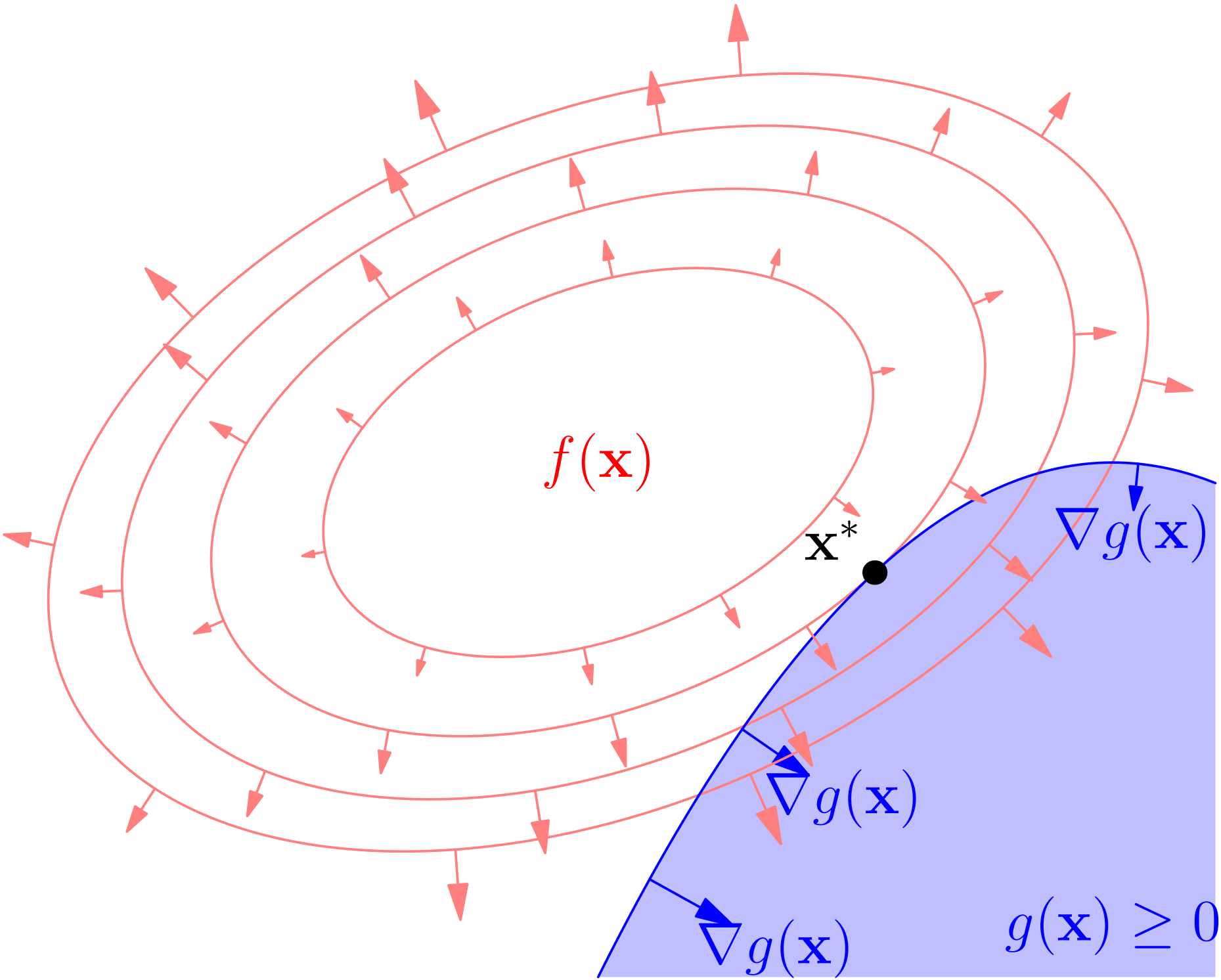
$f(\mathbf{x})$
$\mathbf{x}^*$
$\nabla g(\mathbf{x})$
$\nabla g(\mathbf{x})$
$\nabla g(\mathbf{x})$
$g(\mathbf{x}) \geq 0$